

RETAIL ANALYTICS DECISION SUPPORT SYSTEM

Business Requirements Document

1. Document Information

Item	Details
Document Title	Business Requirements Document
Project	Retail Analytics Decision Support System (RADSS)
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2. Purpose

The purpose of this document is to define the business requirements for the Retail Analytics Decision Support System (RADSS). The system is designed to support retail managers in sales planning, customer value analysis, and product performance evaluation by providing actionable insights from retail transaction data.

3. Business Context

An analysis of the Online Retail II transaction dataset revealed several business patterns that create challenges for retail planning and decision-making.

Sales show a strong seasonal pattern, with revenue consistently peaking in November across both 2010 and 2011 before declining in December and January. This recurring demand fluctuation indicates the need for better sales forecasting and planning.

Customer revenue is highly concentrated, with a small group of customers contributing a disproportionately large share of total sales. In addition, purchase frequency does not consistently reflect customer value, suggesting that revenue contribution and purchasing behavior should be analyzed together.

Product performance is also uneven. Most products generate relatively low sales and revenue, while only a small number of products consistently perform well across both revenue and sales volume. This makes product prioritization and promotional planning difficult without analytical support.

These findings suggest that retail managers require a system that can provide sales forecasting, customer value analysis, and product performance analysis in a unified decision-support environment.

4. Problem statement

Based on the analysis of historical retail transaction data, the business currently faces several decision-making challenges.

PS-01. Seasonal demand planning is reactive rather than predictive.

Recurring sales peaks create pressure on inventory and operational planning, but managers lack a structured mechanism to anticipate demand changes in advance.

PS-02. High-value customers are not systematically identified.

A small segment of customers contributes a large share of revenue, yet the business does not have a consistent method to identify, monitor, and prioritize these customers for retention and marketing activities.

PS-03. Customer value assessment is incomplete.

Purchase frequency alone does not accurately represent customer value, making it difficult to distinguish between frequent low-value customers and high-value strategic customers.

PS-04. Product investment decisions are not sufficiently data-driven.

Most products generate limited commercial impact, while only a small number consistently perform well across both revenue and sales volume. Managers lack an integrated analytical system to prioritize products and optimize promotional investment effectively.

5. Business Objectives

The proposed system aims to achieve the following business objectives:

BO-01. Improve sales planning by providing sales trend analysis and seasonal forecasting.

BO-02. Enhance customer value management by identifying and analyzing high-value customer segments based on revenue and purchase behavior.

BO-03. Improve product performance evaluation by providing integrated analysis of product revenue and sales volume.

6. Stakeholders

The following stakeholders are the primary business users of the proposed RADSS.

Stakeholder	Role	Interest in the System
Sales Manager	Primary user	Monitor sales performance and seasonal trends
Marketing Manager	Primary user	Analyze customer value and customer segments
Product Manager	Primary user	Evaluate product performance and support product decisions
Executive Manager	Decision maker	Review business performance and support strategic planning

7. Scope

7.1. In Scope

The proposed system will provide the following capabilities:

- Sales performance analysis and seasonal sales forecasting.
- Customer segmentation based on revenue and purchase behavior.
- Product performance analysis using revenue and sales volume.

7.2. Out of Scope

The following functions are not included in the initial phase of the system:

- Inventory management and procurement operations.
- Customer relationship management (CRM) functions.
- Online order processing and payment management.
- Automated marketing campaign execution.
- Accounting and financial management.

8. Business Requirements

BR-01. Sales analysis and forecasting

The system shall provide sales analysis and seasonal forecasting capabilities to support sales planning and identify recurring demand patterns.

BR-02. Customer value analysis

The system shall identify and analyze customer segments based on revenue contribution and purchasing behavior to support customer management and marketing decisions.

BR-03. Product performance analysis

The system shall evaluate product performance using sales and revenue data to support product prioritization and portfolio management decisions.

9. Success Criteria

The project will be considered successful if the system enables retail managers to:

- Improve sales planning through seasonal sales analysis and forecasting.
- Identify high-value customer segments more accurately.
- Improve product performance evaluation and prioritization.

10. Assumptions and Constraints

Assumptions

- Historical retail transaction data is available and sufficiently complete for analysis.
- The historical sales patterns are sufficiently representative to support seasonal analysis and forecasting.
- Retail managers have access to the system and can interpret the analytical outputs.

Constraints

- The system is based on historical transaction data and does not include real-time sales data.
- Forecasting accuracy depends on the quality and historical coverage of the available data.
- The project focuses on sales, customer, and product analytics and does not cover full ERP, CRM, or inventory management functionality.